

Opioid, ED Visit, & Mortality

Code ▾

#1. Prepare Data ## 1.1 Activate Packages

Hide

```
install.packages("patchwork")
```

ⓂG2;WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:

https://cran.rstudio.com/bin/windows/Rtools/Ⓜg
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/patchwork_1.3.2.zip'
Content type 'application/zip' length 3352551 bytes (3.2 MB)
downloaded 3.2 MB

package 'patchwork' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
C:\Users\julia\AppData\Local\Temp\Rtmp6rId8S\downloaded_packages

1.2 Merge Datasets

Error in select(., County, year, pop_CDC, Deaths, ER_annual_percapita, :
unused arguments (County, year, pop_CDC, Deaths, ER_annual_percapita, OXYCODONE_annual_percapita, HYD
ROCODONE_annual_percapita, BUPRENORPHINE_annual_percapita, METHADONE_annual_percapita, unemployment_rate,
gini_index, poverty_rate, disability_rate, noinsurance_rate)

1.3 Read Merged Data [Optional]

2. Descriptive Statistics

[[[Table 1 (Part 1)]]]. County Characteristics in Northwestern Alabama, 2016–2019

County <chr>	population <dbl>	pct_gini <dbl>	pct_uninsured <dbl>	pct_unemployee <dbl>	pct_poverty <dbl>	pct_disability <dbl>
Blount	57846	45.7	4.6	3.4	13.6	17.0
Cherokee	25952	45.1	3.5	4.4	13.9	20.2
Colbert	54680	44.7	3.1	4.4	15.5	21.4
Cullman	83109	44.2	4.8	5.1	14.9	20.4
Etowah	102522	45.9	4.8	5.5	17.4	21.2
Fayette	16437	46.1	3.6	6.3	19.4	28.7

County <chr>	population <dbl>	pct_gini <dbl>	pct_uninsured <dbl>	pct_unemployee <dbl>	pct_poverty <dbl>	pct_disability <dbl>	
Franklin	31462	42.8	6.8	4.8	19.4	18.4	
Jackson	51852	45.2	4.3	5.1	18.6	22.0	
Jefferson	659148	50.1	4.7	6.5	16.0	18.5	
Lamar	13878	47.2	6.8	8.8	20.6	25.5	
1-10 of 22 rows 1-7 of 9 columns					Previous	1	2 3 Next

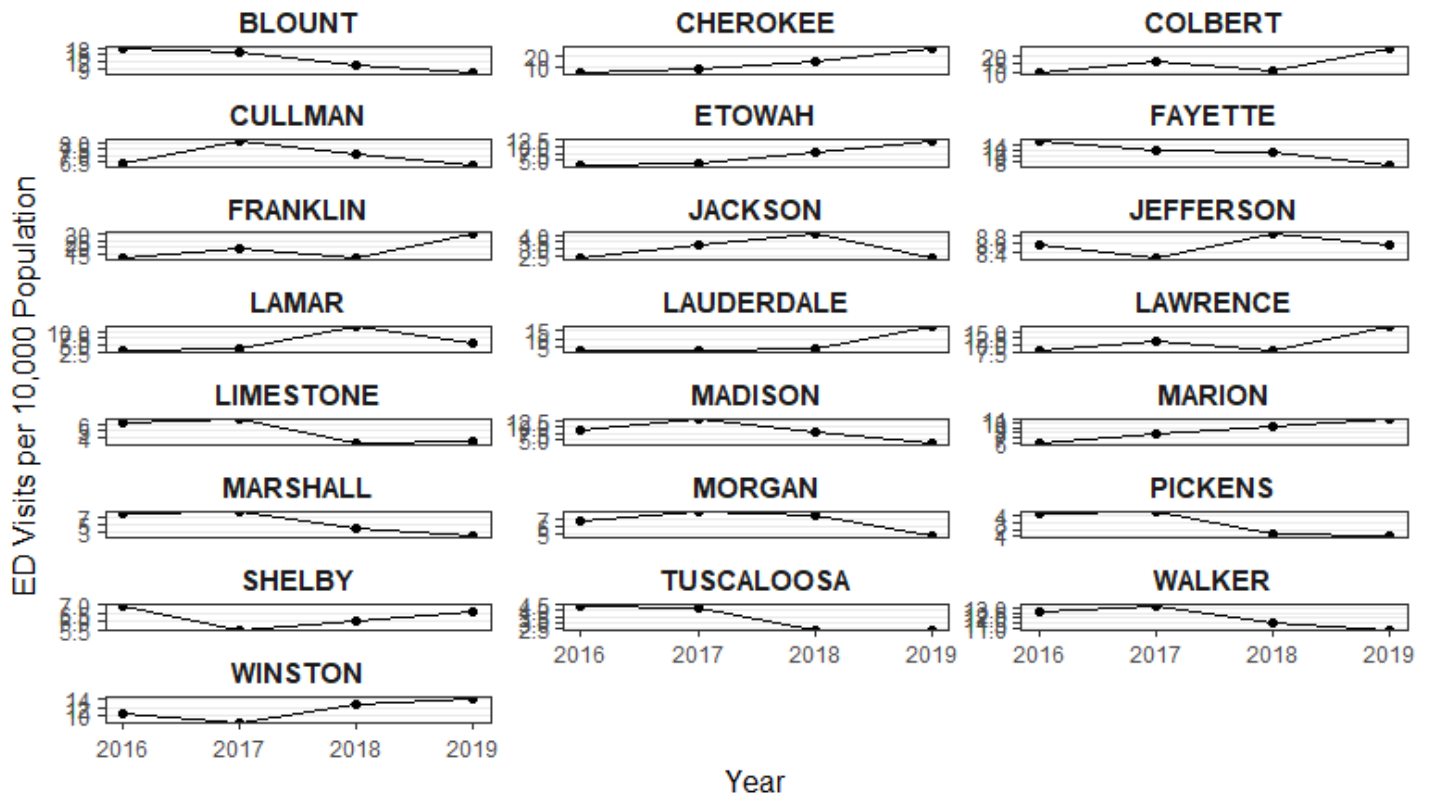
[[[Table 1 (Part 2)]]]. County Characteristics in Northwestern Alabama, 2016–2019

County <chr>	rural <int>	2016_ed_per_10k <dbl>	2017_ed_per_10k <dbl>	2018_ed_per_10k <dbl>	2019_ed_per_10k <dbl>	
BLOUNT	2	17.74	16.27	10.04	7.26	
CHEROKEE	6	3.88	7.36	14.99	28.63	
COLBERT	4	9.54	15.72	11.09	24.26	
CULLMAN	5	6.43	8.09	7.07	6.21	
ETOWAH	4	3.11	3.49	7.80	12.13	
FAYETTE	6	15.09	12.15	10.95	6.75	
FRANKLIN	5	11.70	18.71	11.82	29.97	
JACKSON	5	2.31	3.28	4.07	2.32	
JEFFERSON	2	8.59	8.25	8.84	8.58	
LAMAR	6	2.87	3.60	11.53	5.80	
1-10 of 22 rows 1-6 of 10 columns					Previous	1 2 3 Next

[[[Figure 1]]]. County-Level Emergency Department Visit

Rates, 2016–2019s

County-Level Emergency Department Visit Rates, 2016–2019



3. Imputation and Model Comparison Leave-One-Out Cross-Validation with Truncated Expectation

[!!! Please skip 3.0], we use selected model from Yao's Output

3.0 Comparison All Models [Optional]

Hide

```

getFit <- function(df, df_select_z_int, formula, family = poisson(link = "log"), spatial = "off", spati
otemporal = "off") {

  tr_ex <- function(mu, lower = 1, upper = 9) {
    mu_tr <- sum(((lower:upper) * dpois(lower:upper, mu))) / (ppois(upper, mu) - ppois(lower - 1, mu))
    mu_tr <- ifelse(mu_tr > 9, 9, mu_tr)
    mu_tr <- ifelse(mu_tr < 1, 1, mu_tr)
  }

  tr_ex <- Vectorize(tr_ex, "mu")

  df$Deaths[is.na(df_select_z_int$Deaths)] <- (9+1)/2

  #####

  log_lik_old <- -Inf
  log_lik_new <- Inf
  diff_lik <- Inf
  si <- -1
  tol <- 1e-4

  r <- 0

  previous_fit <- NULL

  mesh <- make_mesh(
    df,
    xy_cols = c("X", "Y"),
    cutoff = diff(range(df$X, na.rm = TRUE)) / 10
  )

  while ((diff_lik > tol) & (si == -1)) {
    r <- r + 1

    cat("r:", r, ", log_lik:", log_lik_old, "\n")
    fit <- try(sdmTMB(
      formula = formula,
      data = df,
      mesh = mesh,
      time = "time_index",
      family = family,
      spatial = spatial,
      spatiotemporal = spatiotemporal,
      offset = df$log_pop,
      previous_fit = previous_fit
    ), silent = TRUE)

    if (class(fit)[1] == "try-error") {
      cat(" sdmTMB fitting failed at iteration", r, "\n")
      break
    }

    previous_fit <- fit
  }

```

```

    if (r == 1) {
      init_fit <- previous_fit
    }

    log_lik_new <- logLik(fit)

    fitted_value <- fitted(fit)

    tmp_fitted_value <- fitted_value[is.na(df_select_z_int$Deaths)]
    tmp_fitted_value <- tr_ex(tmp_fitted_value)

    df$Deaths[is.na(df_select_z_int$Deaths)] <- tmp_fitted_value

    diff_lik <- (log_lik_old - log_lik_new)
    si <- sign(log_lik_old - log_lik_new)
    diff_lik <- abs(diff_lik)

    log_lik_old <- log_lik_new
  }

  list("init_fit" = init_fit, "fit" = previous_fit, "imputed_df" = df)
}

cv_getFit_loo <- function(df, df_select_z_int, formula,
                          family = poisson(link = "log"), spatial = "off", spatiotemporal = "off") {

  n <- nrow(df)

  notna <- which(!is.na(df$Deaths))
  K <- length(notna)

  cv_st <- matrix(NA_real_, nrow = K, ncol = 4)
  colnames(cv_st) <- c("IM_RMSE", "IM_MAE", "EM_RMSE", "EM_MAE")

  r <- 0
  for (k in notna) {
    r <- r + 1
    cat("  Fold", r, "of", K, "\n")
    test_idx <- k
    train_idx <- 1:n
    train_idx <- train_idx[train_idx != test_idx]

    df_train      <- df[train_idx, , drop = FALSE]
    df_select_train <- df_select_z_int[train_idx, , drop = FALSE]
    df_test       <- df[test_idx, , drop = FALSE]

    # Fit model on training data
    fit_k <- try(getFit(df_train, df_select_train, formula,
                       family, spatial = spatial, spatiotemporal = spatiotemporal), silent = TRUE)

    if (class(fit_k)[1] == "try-error") {

```

```

    cv_st[r, "IM_RMSE"] <- NA
    cv_st[r, "IM_MAE"] <- NA
    cv_st[r, "EM_RMSE"] <- NA
    cv_st[r, "EM_MAE"] <- NA
    next
  }

  # Predict on test data
  pred_init <- try(predict(fit_k$init_fit, newdata = df_test, offset = df_test$log_pop, type = "response"), silent = TRUE)

  if (class(pred_init)[1] == "try-error") {
    cv_st[r, "IM_RMSE"] <- NA
    cv_st[r, "IM_MAE"] <- NA
  } else {
    pred_init <- pred_init$est
    # Use only rows with observed Deaths
    pred_notna <- pred_init[!is.na(df_test$Deaths)]
    deaths_notna <- df_test$Deaths[!is.na(df_test$Deaths)]

    rmse <- (((pred_notna - deaths_notna)^2))
    mae <- (abs(pred_notna - deaths_notna))
    cv_st[r, "IM_RMSE"] <- rmse
    cv_st[r, "IM_MAE"] <- mae
  }

  pred <- try(predict(fit_k$fit, newdata = df_test, offset = df_test$log_pop, type = "response"), silent = TRUE)

  if (class(pred)[1] == "try-error") {
    cv_st[r, "EM_RMSE"] <- NA
    cv_st[r, "EM_MAE"] <- NA
  } else {
    pred <- pred$est
    # Use only rows with observed Deaths
    pred_notna <- pred[!is.na(df_test$Deaths)]
    deaths_notna <- df_test$Deaths[!is.na(df_test$Deaths)]

    rmse <- (((pred_notna - deaths_notna)^2))
    mae <- (abs(pred_notna - deaths_notna))
    cv_st[r, "EM_RMSE"] <- rmse
    cv_st[r, "EM_MAE"] <- mae
  }

}

cv_st <- colMeans(cv_st, na.rm = TRUE)
cv_st[c(1, 3)] <- sqrt(cv_st[c(1, 3)])
cv_st
}

```

```
#####
```

```
#####
```

```

cor_df <- df %>% select(
  ER_annual_percapita_z,
  OXYCODONE_annual_percapita_z,
  HYDROCODONE_annual_percapita_z,
  BUPRENORPHINE_annual_percapita_z,
  METHADONE_annual_percapita_z,
  unemployment_rate_z,
  gini_index_z,
  poverty_rate_z,
  disability_rate_z,
  noinsurance_rate_z
)

corrplot::corrplot(cor(cor_df, method = "spearman"))

#####

full_formula <- Deaths ~ ER_annual_percapita_z +
  OXYCODONE_annual_percapita_z +
  #HYDROCODONE_annual_percapita_z +
  BUPRENORPHINE_annual_percapita_z +
  METHADONE_annual_percapita_z +
  #unemployment_rate_z +
  #gini_index_z +
  poverty_rate_z +
  disability_rate_z +
  noinsurance_rate_z +
  rural_factor

fit0 <- getFit(df, df_select_z_int, full_formula)

car::vif(fit0$fit)

#####
#remove the highest VIF until all less than 5
#####

wrapper <- function(X, formulas, df, df_select_z_int) {

  formula <- formulas[[X]]

  cv <- list()

  cv[[1]] <- try(cv_getFit_loo(df, df_select_z_int, formula, spatial = "off", spatiotemporal = "off"),
    silent = TRUE)
  cv[[2]] <- try(cv_getFit_loo(df, df_select_z_int, formula, spatial = "on", spatiotemporal = "off"), s
    ilent = TRUE)
  cv[[3]] <- try(cv_getFit_loo(df, df_select_z_int, formula, spatial = "off", spatiotemporal = "ar1"),
    silent = TRUE)
  cv[[4]] <- try(cv_getFit_loo(df, df_select_z_int, formula, spatial = "on", spatiotemporal = "ar1"), s
    ilent = TRUE)
  cv[[5]] <- try(cv_getFit_loo(df, df_select_z_int, formula, family = nbinom2(link = "log"), spatial =
    "off", spatiotemporal = "off"), silent = TRUE)
  cv[[6]] <- try(cv_getFit_loo(df, df_select_z_int, formula, family = nbinom2(link = "log"), spatial =
    "on", spatiotemporal = "off"), silent = TRUE)

```

```

cv[[7]] <- try(cv_getFit_loo(df, df_select_z_int, formula, family = nbinom2(link = "log"), spatial =
"off", spatiotemporal = "ar1"), silent = TRUE)
cv[[8]] <- try(cv_getFit_loo(df, df_select_z_int, formula, family = nbinom2(link = "log"), spatial =
"on", spatiotemporal = "ar1"), silent = TRUE)

out <- NULL

for (i in 1:4) {
  if (class(cv[[i]]) != "try-error") {
    out <- c(out, cv[[i]])
  } else {
    out <- c(out, rep(NA, 4))
  }
}

out

}

base_vars <- c(
  "ER_annual_percapita_z",
  "OXYCODONE_annual_percapita_z",
  "BUPRENORPHINE_annual_percapita_z",
  "METHADONE_annual_percapita_z",
  "rural_factor"
)

covars <- c("poverty_rate_z", "disability_rate_z", "noinsurance_rate_z", "rural_factor")

make_formula <- function(vars){
  rhs_main <- if(length(vars) > 0){
    paste0("(", paste(vars, collapse = " + "), ") * intervention2017")
  } else {
    "intervention2017"
  }
  rhs_cov <- paste(covars, collapse = " + ")
  as.formula(paste("Deaths ~", rhs_main, "+", rhs_cov))
}

make_formula1 <- function(vars){
  rhs_main <- if(length(vars) > 0){
    paste0("(", paste(vars, collapse = " + "), ") + intervention2017")
  } else {
    "intervention2017"
  }
  rhs_cov <- paste(covars, collapse = " + ")
  as.formula(paste("Deaths ~", rhs_main, "+", rhs_cov))
}

subsets <- unlist(
  lapply(0:length(base_vars), function(k)
    combn(base_vars, k, simplify = FALSE)
  ),
  recursive = FALSE

```

```

)

formulas <- NULL

formulas <- lapply(subsets, make_formula)
names(formulas) <- sapply(subsets, function(v)
  if(length(v) == 0) "none" else paste(v, collapse = "_")
)

formulas1 <- NULL

formulas1 <- lapply(subsets, make_formula1)
names(formulas1) <- sapply(subsets, function(v)
  if(length(v) == 0) "none" else paste(v, collapse = "_")
)

formulas2 <- list()

for (i in 1:length(formulas)) {
  formulas2[[i]] <- formulas[[i]]
  formulas2[[length(formulas) + (i)]] <- formulas1[[i]]
}

nn <- length(formulas2)

formulas2[[nn + 1]] <- Deaths ~ ER_annual_percapita_z
formulas2[[nn + 2]] <- Deaths ~ OXYCODONE_annual_percapita_z
formulas2[[nn + 3]] <- Deaths ~ BUPRENORPHINE_annual_percapita_z
formulas2[[nn + 4]] <- Deaths ~ METHADONE_annual_percapita_z
formulas2[[nn + 5]] <- Deaths ~ poverty_rate_z
formulas2[[nn + 6]] <- Deaths ~ disability_rate_z
formulas2[[nn + 7]] <- Deaths ~ noinsurance_rate_z
formulas2[[nn + 8]] <- Deaths ~ rural_factor
formulas2[[nn + 9]] <- Deaths ~ intervention2017

##### Parallel

nform <- length(formulas2)

cl <- parallel::makeCluster(min(floor(18), nform))
parallel::clusterExport(cl, c("getFit", "cv_getFit_loo", "wrapper"))
parallel::clusterEvalQ(cl, library(dplyr))
parallel::clusterEvalQ(cl, library(readr))
parallel::clusterEvalQ(cl, library(lubridate))
parallel::clusterEvalQ(cl, library(tidyr))
parallel::clusterEvalQ(cl, library(stringr))
parallel::clusterEvalQ(cl, library(janitor))
parallel::clusterEvalQ(cl, library(sdmTMB))
parallel::clusterEvalQ(cl, library(tigris))
parallel::clusterEvalQ(cl, library(sf))

out <- parallel::parLapplyLB(cl, X = 1:nform, fun = wrapper, formulas = formulas2, df = df, df_select_z

```

```
_int = df_select_z_int)

#save(out, file = paste0(repos, "/Data/Yao_out.Rdat"))
```

3.1 The Best Model (From Yao’s output)

X..	County	y...	pop_...	Deaths	ER_annual_percapita	OXYCODONE_annual_percapita
<int>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>
1	BLOUNT	2016	57704	5.608412	0.0017740982	155.9895
2	BLOUNT	2017	58013	5.938220	0.0016266634	140.7245
3	BLOUNT	2018	57840	7.289437	0.0010039639	138.5267
4	BLOUNT	2019	57826	7.111274	0.0007263169	125.0294
5	CHEROKEE	2016	25725	1.338834	0.0003880782	563.6144
6	CHEROKEE	2017	25857	1.277461	0.0007362914	464.7951
7	CHEROKEE	2018	26032	0.000000	0.0014991927	426.5327
8	CHEROKEE	2019	26196	1.433277	0.0028630325	382.8385
9	COLBERT	2016	54216	0.000000	0.0009541810	585.5839
10	COLBERT	2017	54500	0.000000	0.0015723558	466.1062

1-10 of 88 rows | 1-7 of 35 columns

Previous123456...9Next

Hide

fit_imputed

```
Spatial model fit by ML ['sdmTMB']
Formula: formula_imputed
Mesh: mesh (isotropic covariance)
Data: final_fit$imputed_df
Family: gaussian(link = 'identity')

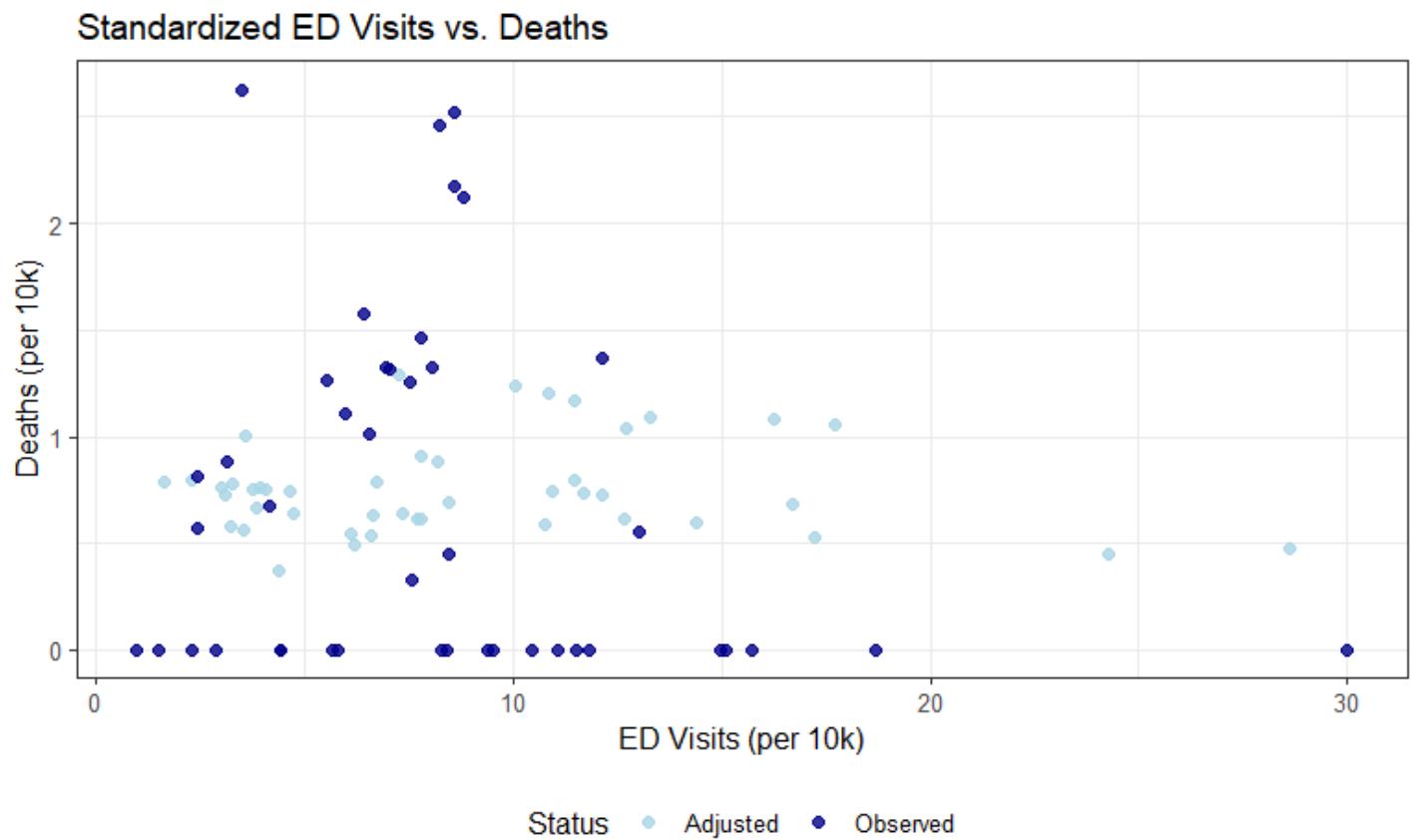
Conditional model:
              coef.est coef.se
(Intercept)      1.87    3.28
ER_annual_percapita_z  -1.48    3.30

Dispersion parameter: 30.79
Matérn range: 2.84
Spatial SD: 0.28
ML criterion at convergence: 426.465

See ?tidy.sdmTMB to extract these values as a data frame.

**Possible issues detected! Check output of sanity().**
```

3.2 Scatted plot: Relationship Between ED Visits and Mortality [Optional]

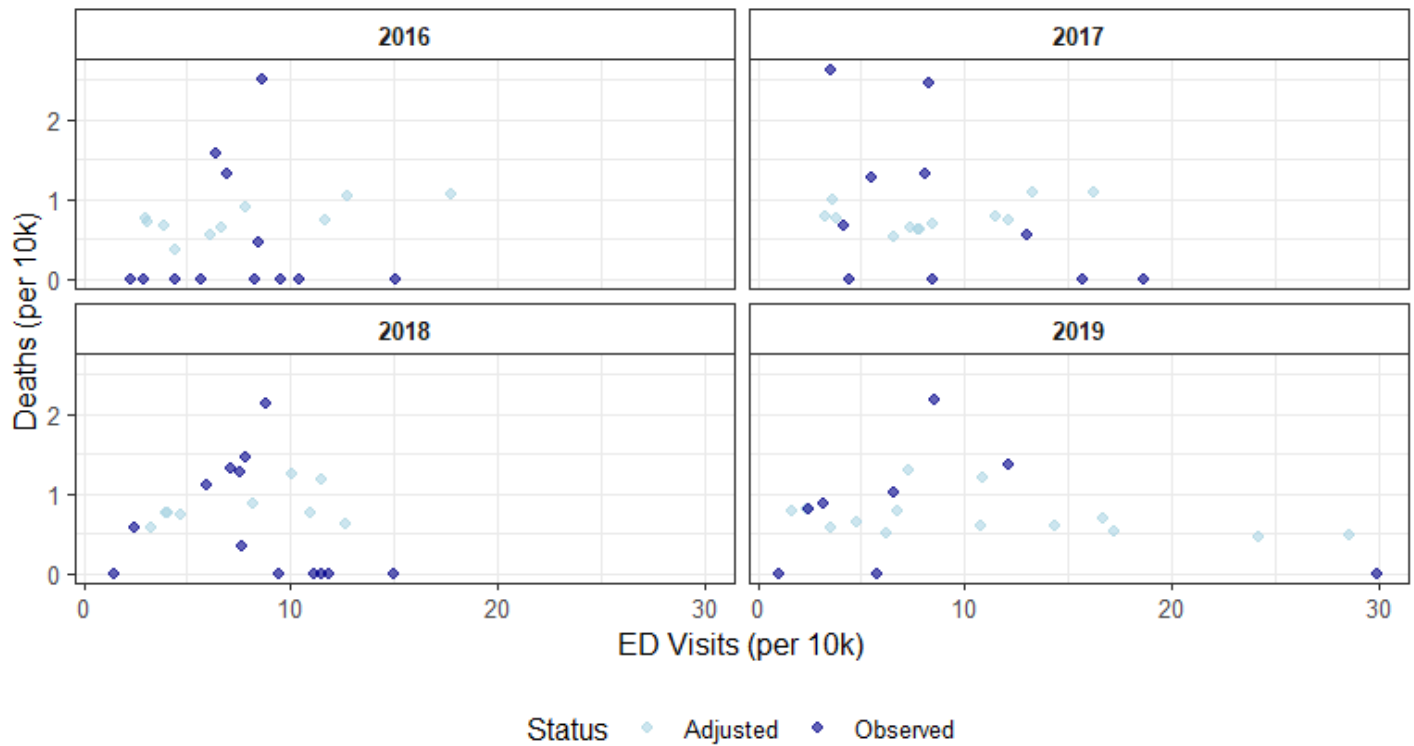


3.3 Scatted plot: (Yearly) Relationship Between ED Visits

and Mortality [Optional]

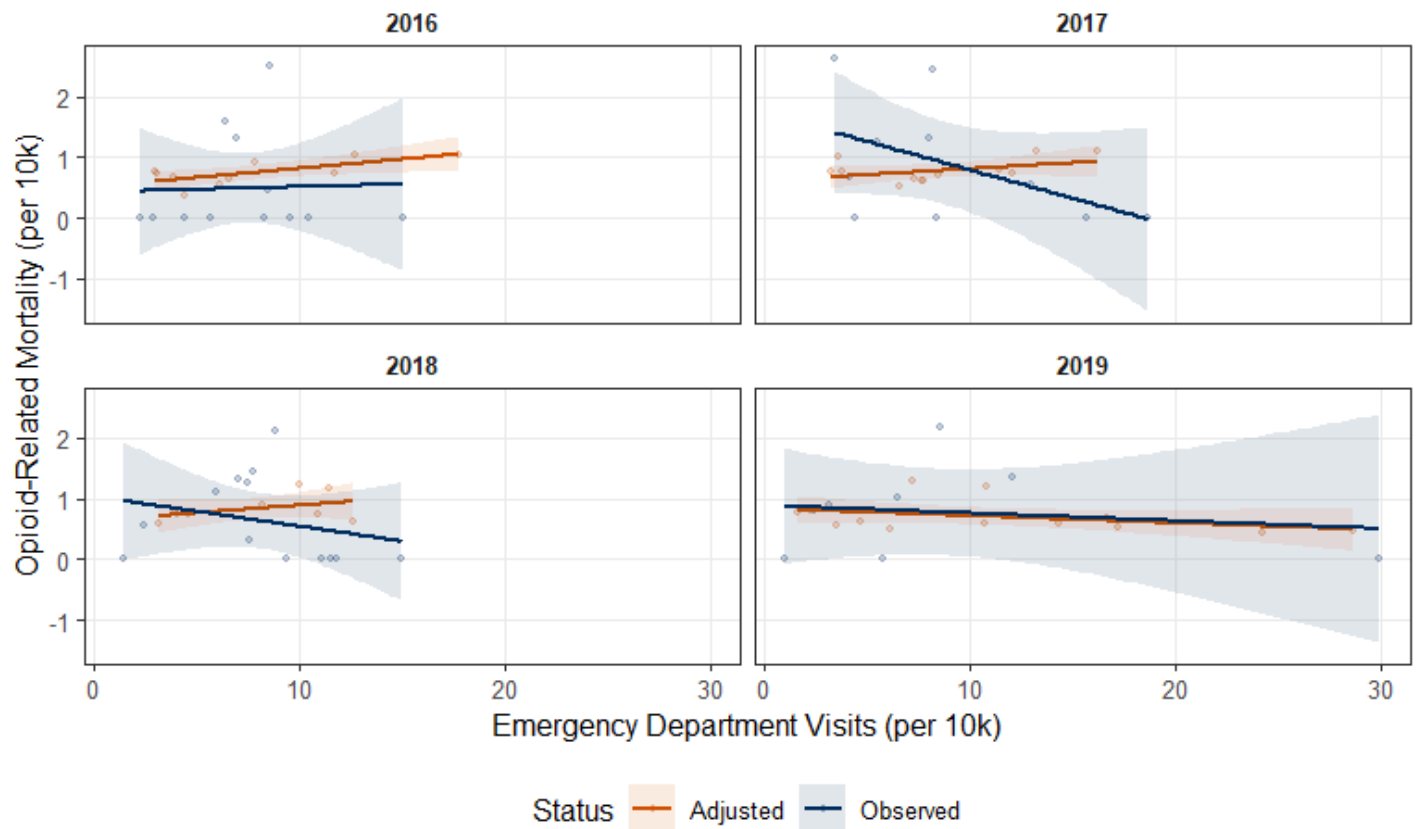
Evolution of Standardized ED Visits vs. Deaths

Comparing Observed and Adjusted Metrics across Years



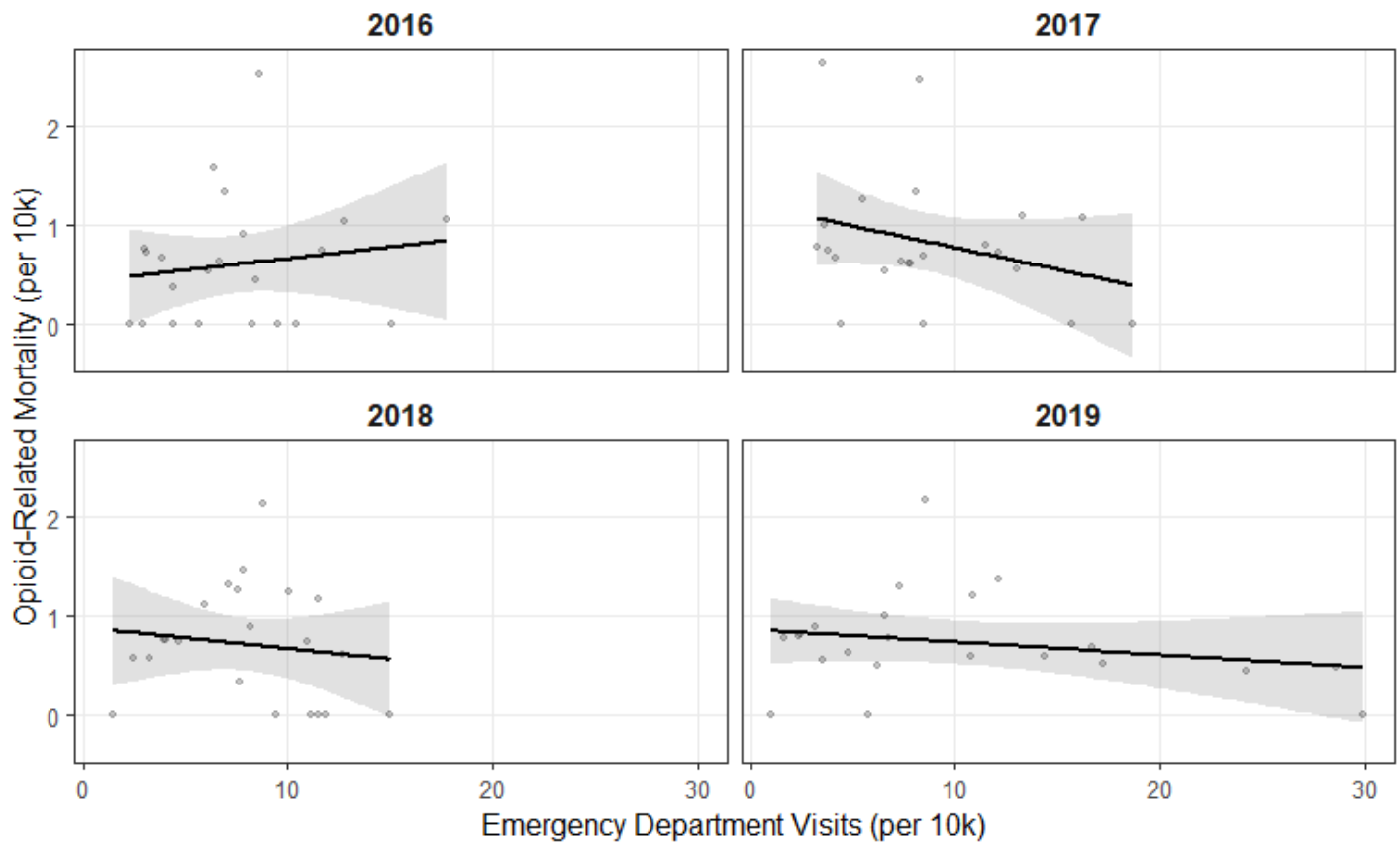
3.4 Scatted plot with lines (seperated by imputated and raw data): (Yearly) Relationship Between ED Visits and

Mortality [Optional]



3.5 Scatted plot with one combined line: (Yearly)

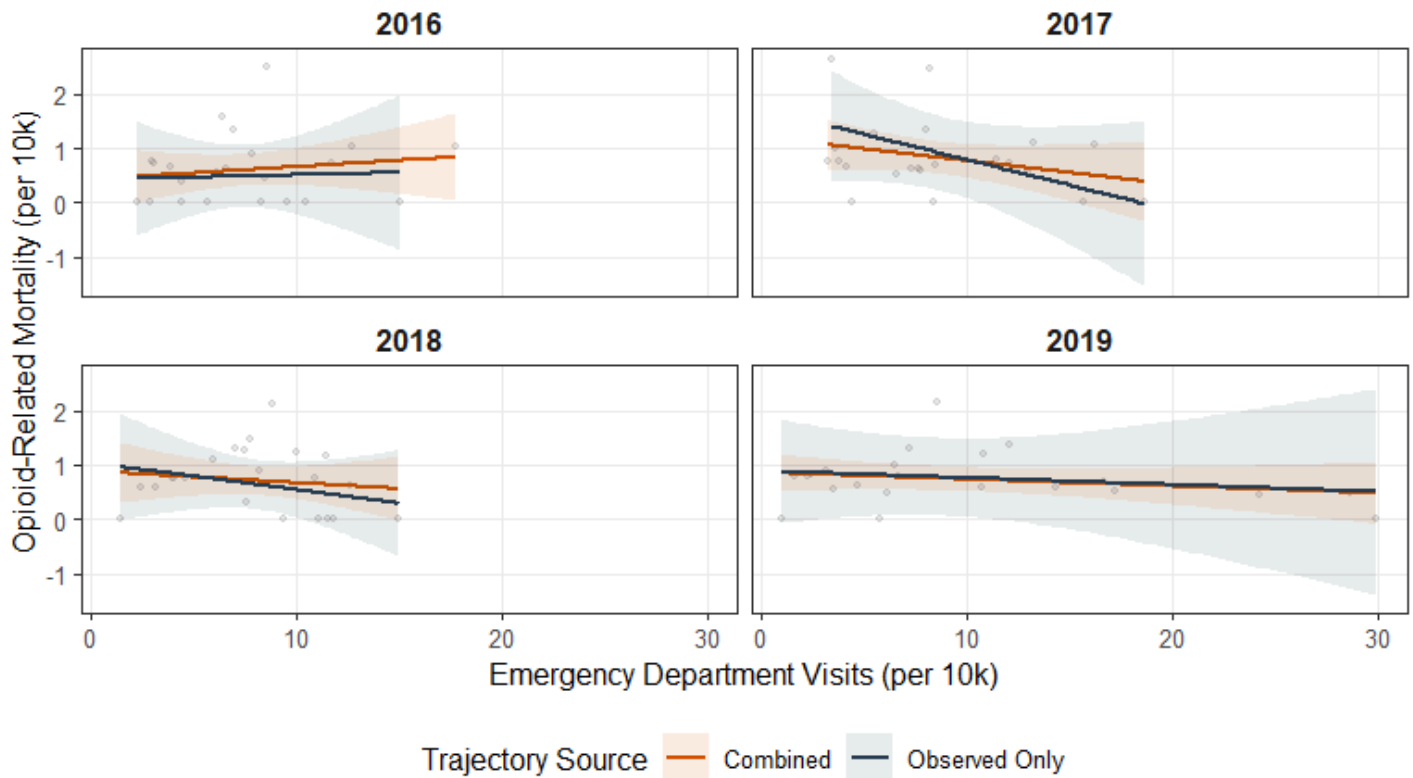
Relationship Between ED Visits and Mortality [Optional]



3.4 Scatted plot with lines (combined and raw data): (Yearly) Relationship Between ED Visits and Mortality

[Optional]

Comparison of Observed and Combined Opioid Mortality Trajectories



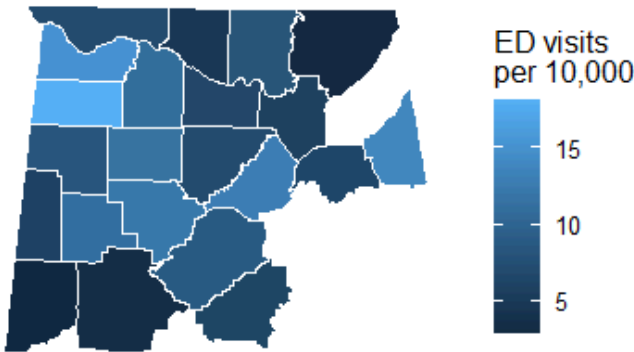
4. “Hot Spot” maps

4.1 [[[Figure 2]]]. Spatial Clustering of ED Visits and Methadone Distribution

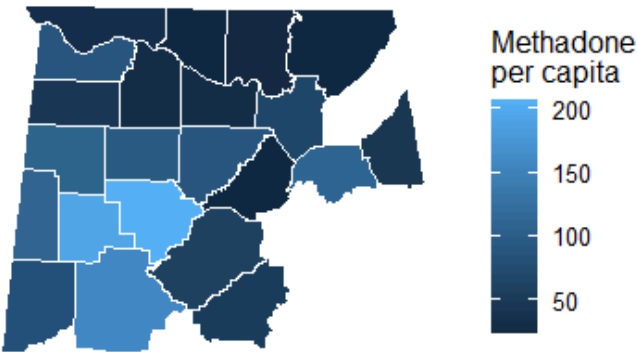
This Figure is newly produced in R. The older one in manuscript was

created by ArcGIS.

A. Opioid-related ED visits



B. Methadone distribution



4.2 Moran’s I Summary Table [Optional]

Variable <chr>	Metric <chr>	2016 <dbl>	2017 <dbl>	2018 <dbl>	2019 <dbl>
Opioid-Related Mortality	Moran's I	0.3126	0.1675	0.3375	0.2606
Opioid-Related Mortality	Expected I	-0.0476	-0.0476	-0.0476	-0.0476
Opioid-Related Mortality	Variance	0.0141	0.0141	0.0158	0.0140
Opioid-Related Mortality	z-value	3.0302	1.8144	3.0626	2.6041
Opioid-Related Mortality	p-value	0.0012	0.0348	0.0011	0.0046
Emergency Department Visits	Moran's I	-0.1179	-0.0998	0.0024	0.2949
Emergency Department Visits	Expected I	-0.0476	-0.0476	-0.0476	-0.0476
Emergency Department Visits	Variance	0.0154	0.0160	0.0163	0.0153
Emergency Department Visits	z-value	-0.5666	-0.4117	0.3915	2.7654
Emergency Department Visits	p-value	0.7145	0.6597	0.3477	0.0028

1-10 of 10 rows

5. Mediation Model

5.1 Imputation

5.5 [Final] Mediator

5.5.0 Drugs Composite

Hide

4. Verify loadings - ensure they are positive (High score = High Burden)
print(pca_opioid\$loadings)

Loadings:

PC1

OXYCODONE_annual_percapita_z0.918

HYDROCODONE_annual_percapita0.908

BUPRENORPHINE_annual_percapita_z0.790

PC1

SS loadings2.291

Proportion Var0.764

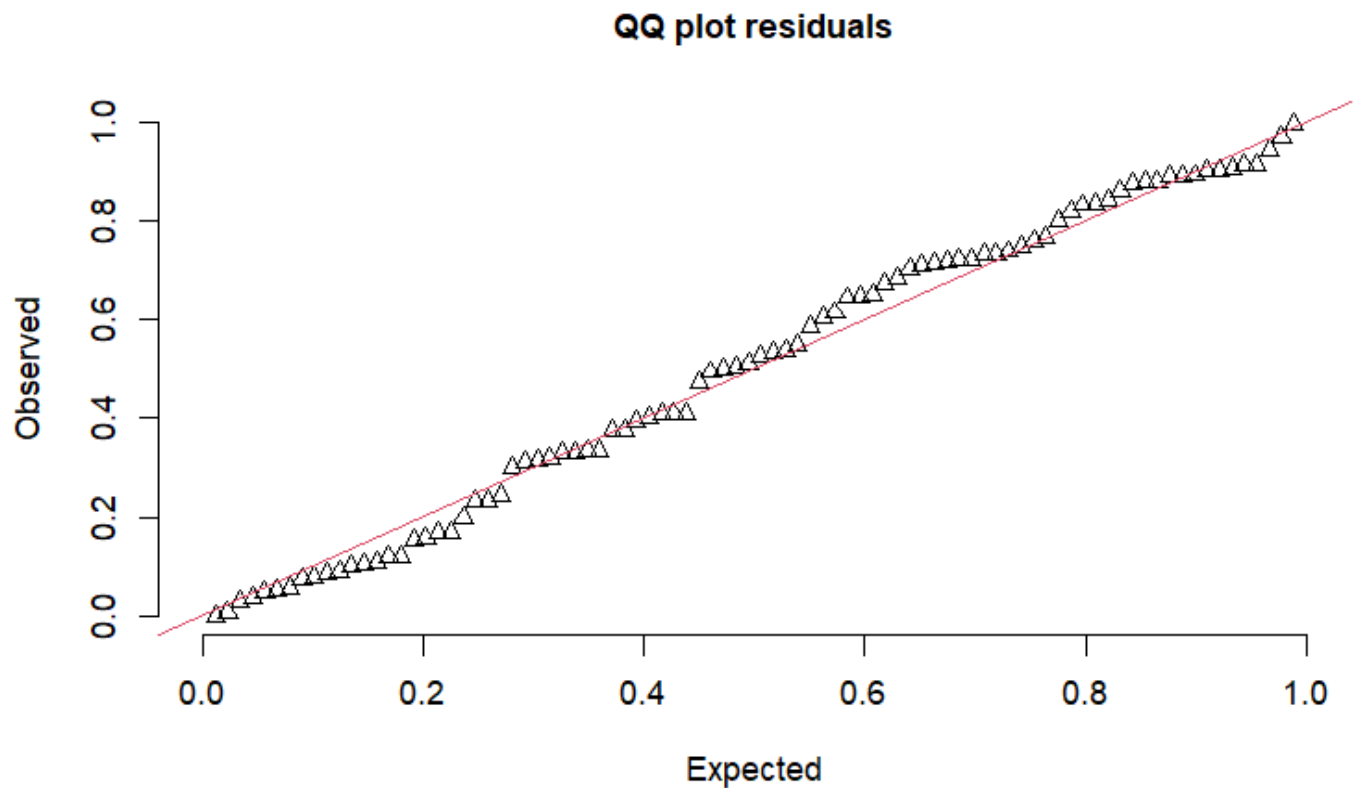
5.5.0 SES Composite

5.5.1 [[[Table 3]]]. Medication-Specific Associations from Spatiotemporal Model

term	Mediator (Rural Base)_Result	Outcome (Mortality)_Result	Mediator (Rural Base)_stars	Outcome (Mortality)_stars
(Intercept)	-5.55 (0.51)	-9.62 (0.38)	***	***
OXYCODONE_annual_percapita_z	-6.45 (2.34)	0.59 (0.17)	**	***
BUPRENORPHINE_annual_percapita_z	1.13 (0.71)	0.25 (0.13)		*
HYDROCODONE_annual_percapita_z	4.09 (1.56)	-1.02 (0.22)	**	***
METHADONE_annual_percapita_z	2.83 (1.34)	0.19 (0.09)	*	*
rural_factor3	-0.27 (1.06)	-0.86 (0.20)		***
rural_factor4	-2.31 (0.44)	0.27 (0.25)	***	
rural_factor5	-3.66 (0.60)	0.70 (0.35)	***	*
rural_factor6	-3.05 (0.57)	-0.94 (0.35)	***	**
gini_index_z	-0.72 (0.16)	0.39 (0.08)	***	***
poverty_rate_z	0.48 (0.26)	-0.03 (0.15)	.	
noinsurance_rate_z	0.17 (0.20)	-0.16 (0.14)		
time_index	-0.03 (0.27)	0.15 (0.25)		
l(time_index^2)	0.15 (0.05)	-0.04 (0.05)	**	
OXYCODONE_annual_percapita_z:rural_factor3	3.39 (2.03)	-	.	
OXYCODONE_annual_percapita_z:rural_factor4	5.80 (1.82)	-	**	
OXYCODONE_annual_percapita_z:rural_factor5	2.94 (2.75)	-		
OXYCODONE_annual_percapita_z:rural_factor6	4.26 (2.00)	-	*	
OXYCODONE_annual_percapita_z:gini_index_z	-0.58 (0.47)	-		
OXYCODONE_annual_percapita_z:poverty_rate_z	-0.89 (0.74)	-		

term	Mediator (Rural Base)_Result	Outcome (Mortality)_Result	Mediator (Rural Base)_stars	Outcome (Mortality)_stars
OXYCODONE_annual_percapita_z:noinsurance_rate_z	-0.24 (0.39)	-		
OXYCODONE_annual_percapita_z:time_index	0.84 (0.42)	-	*	
OXYCODONE_annual_percapita_z:l(time_index^2)	-0.13 (0.07)	-	*	
BUPRENORPHINE_annual_percapita_z:rural_factor3	-0.57 (0.66)	-		
BUPRENORPHINE_annual_percapita_z:rural_factor4	-0.55 (0.41)	-		
BUPRENORPHINE_annual_percapita_z:rural_factor5	-1.38 (0.64)	-	*	
BUPRENORPHINE_annual_percapita_z:rural_factor6	-0.36 (0.49)	-		
BUPRENORPHINE_annual_percapita_z:gini_index_z	0.44 (0.21)	-	*	
BUPRENORPHINE_annual_percapita_z:poverty_rate_z	0.41 (0.23)	-	.	
BUPRENORPHINE_annual_percapita_z:noinsurance_rate_z	-0.40 (0.19)	-	*	
BUPRENORPHINE_annual_percapita_z:time_index	-1.03 (0.27)	-	***	
BUPRENORPHINE_annual_percapita_z:l(time_index^2)	0.15 (0.05)	-	**	
HYDROCODONE_annual_percapita_z:rural_factor3	1.15 (2.15)	-		
HYDROCODONE_annual_percapita_z:rural_factor4	-4.74 (1.40)	-	***	
HYDROCODONE_annual_percapita_z:rural_factor5	-1.92 (2.06)	-		
HYDROCODONE_annual_percapita_z:rural_factor6	-2.45 (1.44)	-	.	
HYDROCODONE_annual_percapita_z:gini_index_z	0.66 (0.37)	-	.	
HYDROCODONE_annual_percapita_z:poverty_rate_z	0.74 (0.74)	-		
HYDROCODONE_annual_percapita_z:noinsurance_rate_z	-0.27 (0.36)	-		
HYDROCODONE_annual_percapita_z:time_index	0.18 (0.35)	-		
HYDROCODONE_annual_percapita_z:l(time_index^2)	0.03 (0.06)	-		
METHADONE_annual_percapita_z:rural_factor3	-4.01 (1.47)	-	**	
METHADONE_annual_percapita_z:rural_factor4	-3.10 (1.22)	-	*	
METHADONE_annual_percapita_z:rural_factor5	-1.59 (1.52)	-		
METHADONE_annual_percapita_z:rural_factor6	-3.52 (1.47)	-	*	
METHADONE_annual_percapita_z:gini_index_z	-0.12 (0.38)	-		
METHADONE_annual_percapita_z:poverty_rate_z	1.07 (0.31)	-	***	
METHADONE_annual_percapita_z:noinsurance_rate_z	-0.76 (0.22)	-	***	
METHADONE_annual_percapita_z:time_index	0.19 (0.18)	-		
METHADONE_annual_percapita_z:l(time_index^2)	-0.01 (0.03)	-		
ER_annual_percapita_z	-	0.83 (0.45)		.
ER_annual_percapita_z:rural_factor3	-	-0.37 (0.30)		
ER_annual_percapita_z:rural_factor4	-	-0.17 (0.31)		
ER_annual_percapita_z:rural_factor5	-	-0.20 (0.45)		
ER_annual_percapita_z:rural_factor6	-	-0.16 (0.41)		
ER_annual_percapita_z:gini_index_z	-	0.09 (0.15)		
ER_annual_percapita_z:poverty_rate_z	-	0.12 (0.14)		
ER_annual_percapita_z:noinsurance_rate_z	-	-0.07 (0.14)		
ER_annual_percapita_z:time_index	-	-0.53 (0.38)		
ER_annual_percapita_z:l(time_index^2)	-	0.09 (0.07)		

5.5.2 Adjusting for Distributional Assumptions (Overdispersion)



6. Shiny

[Hide](#)

```
# 4. Save this object specifically for the Shiny App
save(final_fit, df, file = "best_model.RData")

load(paste0(repos, "/Data/best_model.RData"))
```